

7. (a) A class investigates radioactive decay. They model decay using **8-sided** dice.



Each of the 10 groups has 500 dice.
They throw the dice and remove all which land with an 8 facing upwards.
These represent decayed nuclei.
They repeat 7 more times and record the number of dice remaining after each throw.
Each group's results are then added together.

- (i) Freya suggests that it is good practice to add the results together to give a larger sample size.
Explain whether you agree. [1]

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- (ii) The teacher calculates that after the first throw **around** 440 of the 8-sided dice should remain out of the 500.
Explain how she determined this number. [2]

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- (b) The results from the experiment are given in the table below.

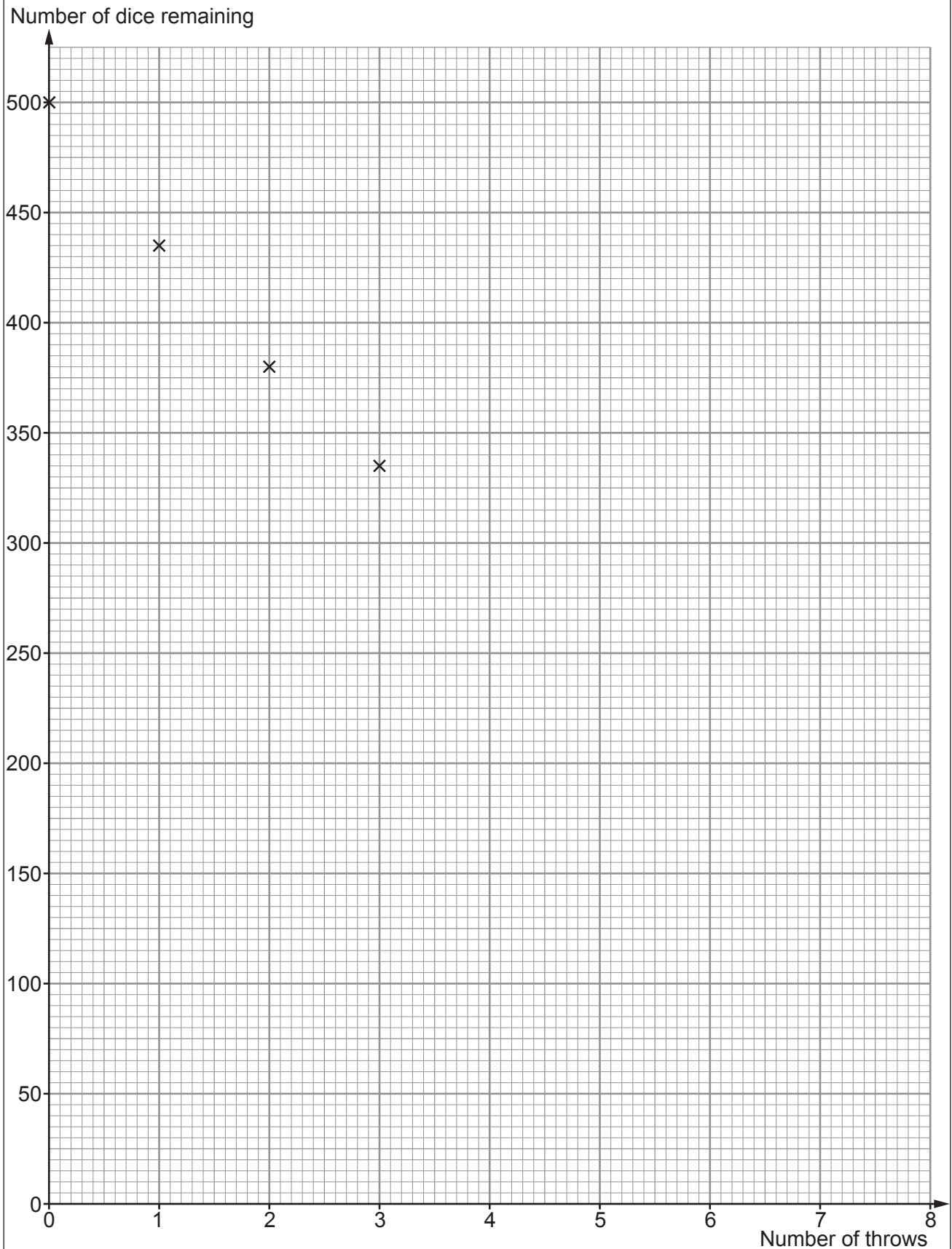
Number of throws	Number of dice remaining
0	500
1	435
2	380
3	335
4	291
5	256
6	224
7	196
8	171



- (i) Plot the data on the grid below and draw a suitable curve.
The first 4 points have been plotted for you.

[3]

Examiner
only



Examiner
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- (ii) **Add lines to the graph** to find the number of throws required to halve the number of dice. This is the half-life.
Give your answer to 1 decimal place. [2]

number of throws =

- (iii) The experiment is repeated with 10-sided dice.
leuan suggests the data could be used to model nuclear decay with a shorter half-life than with 8-sided dice.
Explain whether you agree. [2]

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END OF PAPER

